MIT RES.6-012 L05.11 Linearity of Expectations

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$$

Proof

Let

$$Y = g(X) = aX + b.$$

Assume X is a discrete random variable with probability mass function $P_X(x)$. Then

$$\begin{aligned}
\mathbb{E}[Y] &= \sum_x g(x)P_X(x) \\
&= \sum_x (ax + b)P_X(x) \\
&= \sum_x axP_X(x) + \sum_x bP_X(x) \\
&= a\sum_x xP_X(x) + b\sum_x P_X(x) \\
&= a\mathbb{E}[X] + b(1) \\
&= a\mathbb{E}[X] + b.
\end{aligned}$$

Therefore,

$$\boxed{\mathbb{E}[aX + b] = a\mathbb{E}[X] + b.}$$

Connection to $g(\mathbb{E}[X])$

Since $g(x) = ax + b$, we have

$$\mathbb{E}[g(X)] = a\mathbb{E}[X] + b.$$

Also,

$$g(\mathbb{E}[X]) = a\mathbb{E}[X] + b.$$

Therefore,

$$\boxed{\mathbb{E}[g(X)] = g(\mathbb{E}[X])}$$

for linear functions of the form

$$g(x) = ax + b.$$